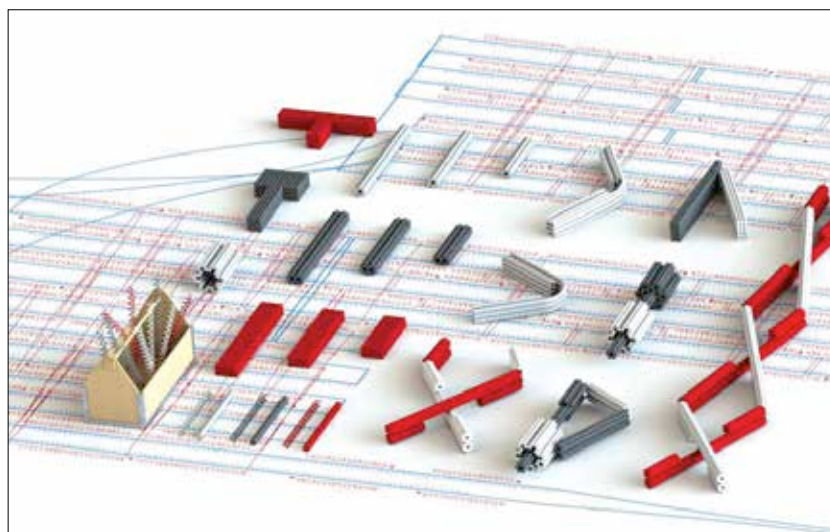


degrees celsius for 10 minutes, and a liquid or gas added (usually hexamethyldisilazane, or HMDS for short) which sticks to the wafer, forming a thin layer on top of the silicon dioxide that is water repellent. Next, a thin layer of photoresist is applied and the wafer is then spun at high speeds, so as to thin out this photoresist layer to the desired thickness. This process is called spin coating, and you can find some interesting DIY experiments on YouTube of people building one at home using old hard drives! The excess photoresist which has not bonded to the silicon dioxide is then baked off the wafer at 100 degrees C.

The next step uses a 'mask' and high intensity UV light or a laser (depending on the wavelength needed) to do the actual lithography, which is akin to pressing the shutter button on a film camera. Like in a camera, here also lenses are used to focus the high intensity light.

The next step is to 'develop' the layer, much like a film photographer would develop film in a dark room. In fact the next liquid layer that's added to aid in the process is called the 'developer', and it too is spin coated on to the wafer. Depending on the type of photoresist and developer used, the shape of the mask is either washed away or kept. Positive photoresist, when exposed to the high intensity light, becomes soluble in the developer, and is washed away. This is the most common type of developer and photoresist. Negative photoresist means the parts of the layer not exposed to the high intensity light will be washed away. This is a rarer use case, but it all depends on how easy the mask is to create – some-



DNA Origami Mechanisms

Credit: Royal Society of Chemistry

times it's easier to make a mask for negative photoresists.

An additional step called etching can be done here. This is basically a removal of the base silicon layer, using the same mask. Finally, the photoresist itself has to be removed, which is done using a solvent called a 'stripper' (such as when you strip the paint off something). And that's how one layer of chip geometry is done. Modern systems stack geometries, so this whole process is repeated until all desired layers have been stacked one on top of another, and the chip is complete.

Electron beam: With electron beam lithography, the name pretty much gives it all away. Yes, instead of light or lasers, a beam of electrons is used to draw on a layer called a resist. The rest of the procedure is pretty much similar to photolithography. The biggest advantage of this method is that it has a much higher resolution than optical nanolithography, and can draw patterns with a resolu-

tion of just under 10 nm, and this is because there is no mask involved... the electron beam itself is moved around just as you would a pencil, and the precision of beam movement plays a role in deciding the resolution. It's not the only factor, because being electrons, they're particles with mass, and suffer from a scattering effect. When shooting a beam of electrons, some get absorbed temporarily into the atoms or molecules, giving rise to ions, while others might be reflected at various angles through the substrate. How wide the angle of this scattering also decreases the resolution of the system (increases nm size of the channels being written). In some instances the scattered electrons can ruin the layer by exiting too close to another drawn channel, and this is what results in wastage.

Nanoimprint: This is a method that uses an imprint layer to mechanically alter the resist layer. Thus it's like a stamper that uses a mold to stamp a shape into the resist layer,

Molecular printing

It is expensive to create DNA structures, but the new approaches in molecular printing may drastically bring down the cause, and the self replicating nature of the molecule can be put to use as well.
<http://dgit.in/nanoprint>

Future production

A broad review of future micro, nano and even femto scale manufacturing processes and a peek into the potential of Industry 4.0.
<http://dgit.in/nanofemto>

Laser manufacturing

Find out how nanotubes, nanowires and nanoparticles are produced using cutting edge laser technologies, through the process of additive manufacturing.
<http://dgit.in/nanolaser>